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Charting the circuit QED design landscape using optimal control theory

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With recent progress in qubit coherence, superconducting quantum circuits have become a strong candidate for quantum computing. They can be designed to implement a wide range of operations, but also present an inherent trade-off between scaling and having good control over the system. In this paper, we systematically explore the landscape of possible gate sets and their implementation using a wide range of optimal control theory. We identify an optimal operating region outside of the usual experimental design space, where multiple orders of magnitude increase in efficiency are observed, which we name the ‘green superconducting qubit regime’. An additional point in this region, a universal gate set, is reached by applying microwave fields for gate duration of 10 ns, which would approach the limit of lifetime-limited coherence. Our systematic quantum control control approach is easily adapted to explore the parameter landscape of other quantum technology candidates.

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INTRODUCTION

For quantum computing to achieve its full potential, efficient operations become vital to success. This involves all relevant steps for operating a device, such as state preparation of quantum gate implementations. One such aspect is the temporal ‘lifetime’ of the qubit, that determines the allowed possible duration to carry out the task at hand.^{1–3} Quantum control theory (QCT)^{4–6} has proven itself a valuable tool for identifying these performance targets. Typical control tasks include the preparation of non-classical states, as shown in, say, combined with a ‘green superconducting’⁷ or ‘the quantum advantage’^{8,9} era. This is done by controlling the system and its environment, to compensate for decoherence and relaxation. Here, these tools have been optimized for hardware parameters of the respective system. While we have a fully featured QCT approach for gate synthesis and, as the most advanced control techniques, they may not be the most versatile tool for the physical system at hand. In the end, we consider the task of isolating the desired qubit, removal of all gates for two superconducting qubits (see Fig. 1) within the lifetime of excited lifetimes.

Superconducting circuits are the one of the most promising architectures for near-term computing tasks. The development of circuit QED¹⁰ is a solid platform for quantum technology, it provides a natural environment for quantum control. The development of circuit QED¹⁰ is a solid platform for quantum technology, it provides a natural environment for quantum control. The development of circuit QED¹⁰ is a solid platform for quantum technology, it provides a natural environment for quantum control.

First, principal operations have been used to create quantum computing ability via a ‘green superconducting’⁷ or ‘the quantum advantage’^{8,9} era. This is done by controlling the system and its environment, to compensate for decoherence and relaxation. Here, these tools have been optimized for hardware parameters of the respective system.

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In this work, we explore a wide range of possible gate sets and their implementation. We identify an optimal operating region outside of the usual experimental design space, where multiple orders of magnitude increase in efficiency are observed, which we name the ‘green superconducting qubit regime’. An additional point in this region, a universal gate set, is reached by applying microwave fields for gate duration of 10 ns, which would approach the limit of lifetime-limited coherence. Our systematic quantum control control approach is easily adapted to explore the parameter landscape of other quantum technology candidates.

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